

EX NO:2

Chebyshev IIR Filter

2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

Objectives:

1. To design and analyze the magnitude response, phase response, impulse response, and group delay characteristics of the Chebyshev Type-I IIR low-pass filter for the given passband and stopband specifications.
2. To analyze the impulse response, and group delay characteristics of the Chebyshev Type-I IIR low-pass filter for the given passband and stopband specifications.
3. To examine the pole-zero plot and verify the stability and performance of the designed low-pass filter.
4. To determine the minimum filter order and cutoff frequency required to satisfy the design constraints.

Procedure

1. Open MATLAB and create a new script file.
2. Define the filter specifications such as passband ripple (R_p), stopband attenuation (R_s), passband frequency (F_p), stopband frequency (F_{sb}), and sampling frequency (F_s) for the Chebyshev Type-I filter design.
3. Normalize the passband and stopband frequencies with respect to the Nyquist frequency ($F_s/2$).
4. Use the `cheb1ord()` function to determine the minimum order and cutoff frequency of the Chebyshev Type-I low-pass and high-pass filters.
5. Design the Chebyshev Type-I low-pass and high-pass filters using the `cheby1()` function and obtain the filter coefficients.
6. Display the filter order and cutoff frequency, and compute the frequency response of the designed filters using the `freqz()` function.
7. Plot the magnitude response, phase response, impulse response, group delay response, and pole-zero plot using the appropriate MATLAB functions to analyze the filter characteristics and stability.

8. Execute the program, analyze the obtained responses and filter characteristics, record the results, and verify that the designed Chebyshev Type-I filters satisfy the given specifications

Program (MATLAB Code):

```
clc; clear;
```

```
Wp=0.4; Ws=0.55; Rp=1; As=40;
```

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Filter using MATLAB

```
[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
```

```
[b,a]=cheby1(N,Rp,Wn);
```

```
fprintf('Filter Order = %d\n',N);
```

```
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
```

```
if all(abs(roots(a))<1)
```

```
disp('System is STABLE');
```

```
else
```

```
disp('System is UNSTABLE');
```

```
end
```

```
% ----- GRAPHS -----
```

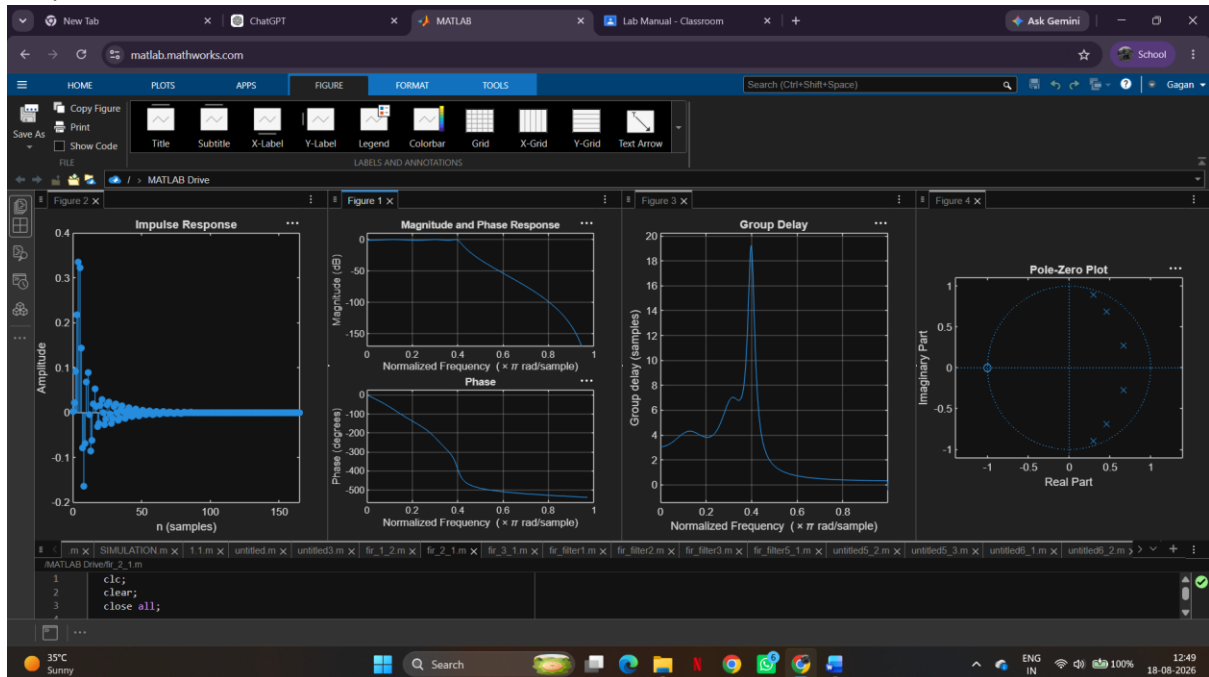
```
figure; freqz(b,a); % Magnitude + Phase
```

```
figure; impz(b,a); % Impulse Response
```

```
figure; grpdelay(b,a); % Group Delay
```

```
figure; zplane(b,a); % Pole-Zero
```

Output:



Filter Order = 6

Cutoff Frequency = $0.40 \times \pi$ rad/sample

System is STABLE

Result:

1. The Chebyshev Type-I IIR low-pass filter was successfully designed, and its magnitude response, phase response, impulse response, and group delay were analyzed.
2. The impulse response and group delay characteristics were obtained and found to satisfy the given specifications.
3. The pole-zero plot confirmed that all poles lie inside the unit circle, verifying the stability and proper performance of the filter.
4. The minimum filter order and cutoff frequency were successfully determined and found to satisfy the given design specifications.